Meeting 1 : planning + short idea of shielding

→ everything falls back to maxwell

Meeting 2 : Literature

→ intro shielding
  → what ?
  → why ?
  → how ?
    → why like that ?

Ampère : $\vec{\nabla} \times \vec{B} = \vec{J} + \mu_0 \varepsilon_0 \cdot \frac{\partial \vec{E}}{\partial t}$: changing E ⇒ some current, leads to a moving magnetic field

Faraday $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$: changin B = Φ leads to E moving = some current

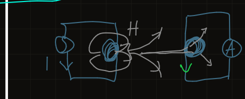
Gauss $\vec{E}$ $\vec{\nabla} \cdot E = \rho / \varepsilon_0$: the density is proportional to the amount of emitted field

Gauss $\vec{B}$ $\vec{\nabla} \cdot B = 0$: all magnetic fields must close ⇒ magnetic monopoles do not exist

Meeting 2 : Intro to shielding

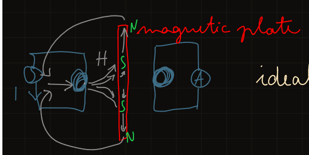
1 : What is shielding ? (yt + Celozzi)

* Reducing or completely blocking EM fields (using conductive or magnetic planes)

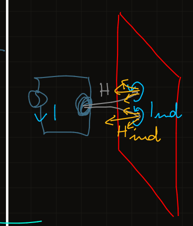


the field going throug induces a current

→ Faraday's Law : $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$: a changing magnetic field will move the electric field : E moving ⇒ Q moving = current



ideally nothing gets through so no induction happens



ideally all incoming $\vec{H}$ induces $\underbrace{\text{eddy currents}}_{}$ with R → 0 for perfect dipoles

$J = \sigma \cdot E$. ohm

Again Faraday : $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Lenz

↳ opposite direction ⇒ eddy currents the other way ⇒ Hind the other way → $\vec{H}$ cancels

2 : Why use shielding (Celozzi)

* Improving EMC
  → performance of electronics
    → small to complex devices and systems
  → public health : signs about radiation
    : people with electronics in them
  → communication security
  → decrease coupling

3 : How do we shield ? (yt + Celozzi

*P 18 : Continuity ?

$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$ — not ρ as conductivity ⇒ charge density

$\vec{\nabla} \times \vec{H} = \vec{J} + \cancel{\mu_0} \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$ no $\vec{B}$

↻ doesn't diverge

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \left(\vec{J} + \cancel{\mu_0} \varepsilon_0 \cdot \frac{\partial \vec{E}}{\partial t}\right)$$

$$0 = \vec{\nabla} \cdot \vec{J} + \cancel{\mu_0} \cdot \varepsilon_0 \cdot \frac{\partial}{\partial t}\left(\vec{\nabla} \cdot \vec{E}\right)$$

$$= \vec{\nabla} \cdot \vec{J} + \cancel{\mu_0} \cdot \varepsilon_0 \cdot \frac{\partial}{\partial t} \rho / \varepsilon_0 = \vec{\nabla} \cdot \vec{J} + \cancel{\mu_0} \cdot \frac{\partial \rho}{\partial t} \quad \square$$

⇒ if current flows out then the $\underbrace{\text{concentration}}_{\text{density}\downarrow}$ of charges $\underline{\text{decreases}}$ $= -\frac{\partial \rho}{\partial t}$

IMPLIES CONSERVATION OF CHARGE

* how is a shield characterized

→ shielding effectiveness : SE $= 20 \log_{10}\left(\frac{H_{ext}}{H_{int}}\right)$ — what goes through — what is emitted

→ skin depth : $\delta = \sqrt{\frac{2}{\omega \cdot \mu \cdot \sigma}} = \sqrt{\frac{1}{\pi \cdot f \mu \cdot \sigma}}$: f↑ smaller depth → eddy currents get pushed to the surface ⇒ J↑↑

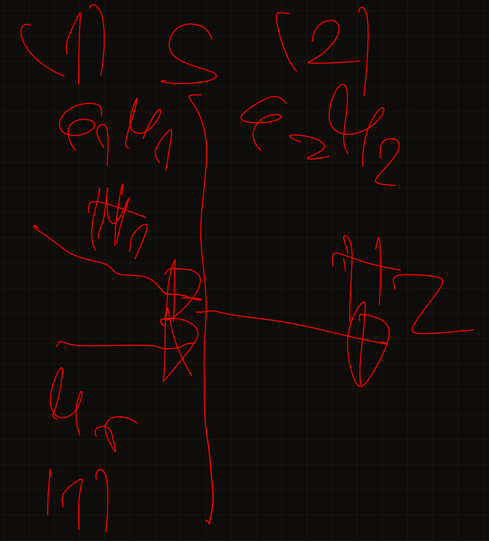
↳ better conductivity
= more J possible
⇒ even more concentration
⇒ even better shielding

Lenz : J↑↑ $\xrightarrow{ind}$ H_{ind}↑↑

very strong opposite field
→ cancels out or can reflect

→ once the field gets through & it's intensity decreases exponentially

next : SE models
→ some matlab



$$\boxed{n_n \times (H_1 - H_2) = j_s}$$

$$\boxed{H_1 = H_n + j H_t}$$

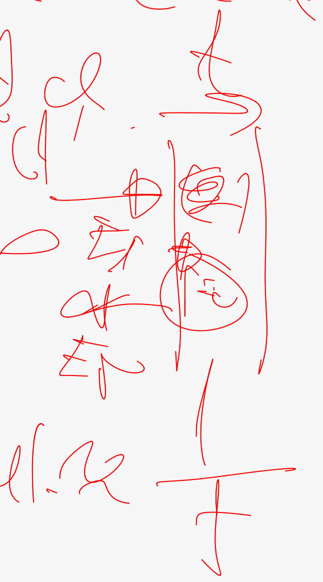
$$e_1 = e_n + j e_t$$

$$\left.\begin{aligned} e_1^n &= e_2^n \\ \mu_1^n \cdot H_1 &= \mu_2^n H_2 \end{aligned}\right\}$$

$$\left.\begin{aligned} H_1^t &= H_2^t \\ e_1^t \epsilon_1 &= e_2^t \cdot \epsilon_2 \end{aligned}\right\}$$

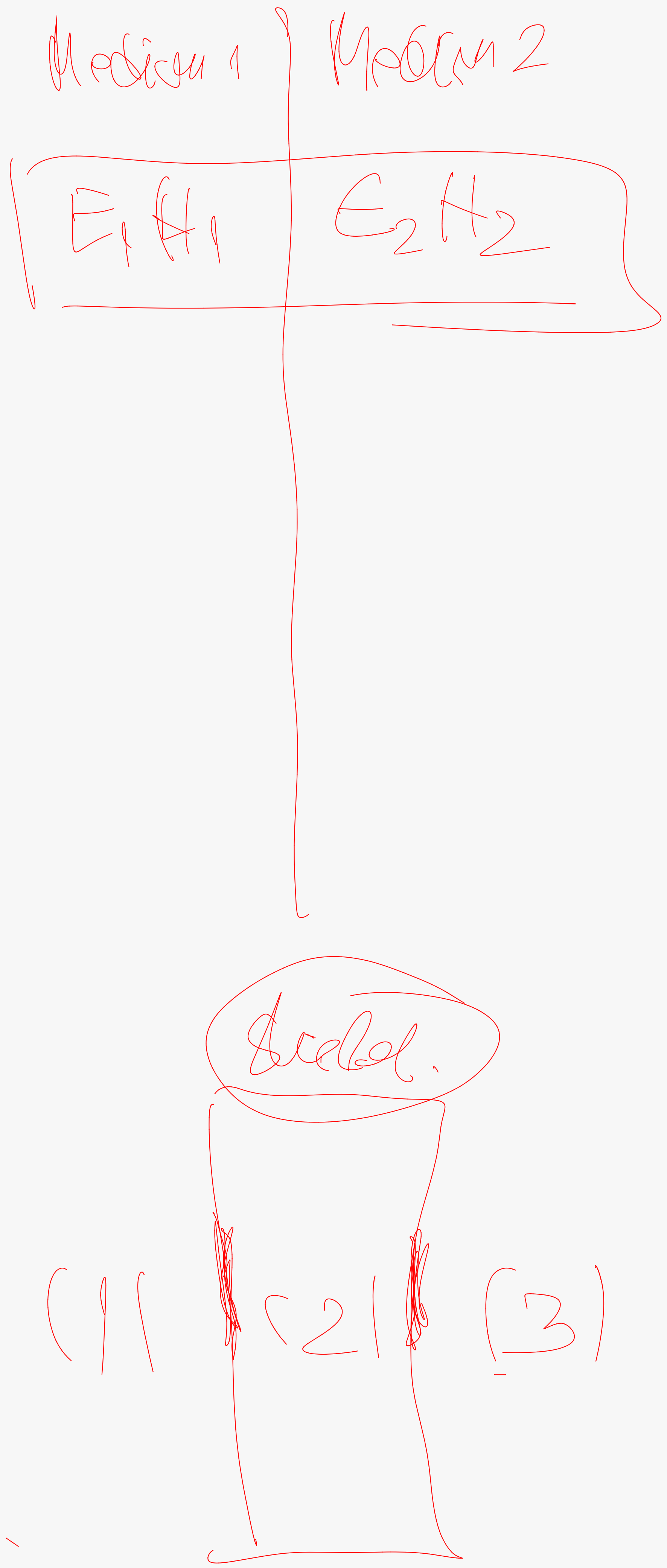
PEC - Perfect electric conductor

$\sigma \longrightarrow \infty$



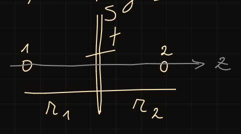
δ (skin depth) ε

$$\boxed{\delta = 0}$$


Medium 1
Medium 2
$E_1 H_1$
$E_2 H_2$
Stiebel.
(1)
(2)
(3)

MOSER

1) Voltages



induced in 1 by eddy in S

$$V_1' = \underbrace{M(2\cdot r_1)\, I\cdot\omega\cdot\cos(\omega t)}_{\text{direct induction}} + \underbrace{\int_{2r_1}^{\infty} M'(z)\cdot I\cdot\omega\cdot\cos(\omega)\cdot\overbrace{\left(t-\frac{z-2r}{v}\right)}^{\text{delay}}\, dz}_{\text{after reflection}}$$

reflected; $2\pi f$; current in loop; magnetization

after reflection → everything that missed

V_2' = very ugly exponential ...

→ dependent on all parameters

if $\frac{r_1}{a} < 0{,}9$, $> 0{,}15$

V_2'' : no shield $= -j\omega I\cdot M_{1\to 2}$

$V_2 = V_2' + V_2''$

↳ insertion loss $= \left|\frac{\text{zonder}}{\text{met + zonder}}\right| = \left|\frac{V_2''}{V_2}\right|$

$$= \sqrt{(a\cdot t\cdot\sigma_t\cdot f/78{,}8)^2 + 1}$$

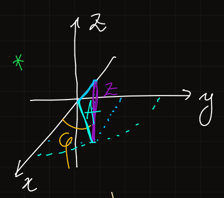
$$S = 20\log\left(\left|\frac{V_2''}{V_2}\right|\right) = 10\log\left((a\cdot t\cdot\sigma_t\cdot f/78{,}8)^2 + 1\right)$$

↳ not dependent on $r_1 + r_2$ and μ

2) Wave eqn:

$$\overset{\nabla^2}{\Delta}\vec{A} + k_0^2\vec{A} = 0 \quad (1)$$

using cylindrical coordinates*: bessel wave surfaces



length in same direction φ

So: $\vec{A}$: Amplitude $= A_\varphi\cdot\vec{\varphi}$ ↳ angle φ → how the current flows

→ no current in r or z ⇒ $A_r = A_z = 0$

thus: $\nabla\cdot\vec{A} = 0 + \boxed{\frac{1}{r}}\frac{\partial A_\varphi}{\partial\varphi} + 0$ → 0 : cylindrical sym = turning has no effect

$= 0$

(greens: $\int_0^\infty e(-\tau_0 z)\, J_0(\lambda a)\frac{\lambda}{\tau_0}\, d\lambda$

$$\nabla\times\vec{A} = -\frac{\partial A_\varphi}{\partial z}\vec{r} + \left(\frac{\partial A_\varphi}{\partial r} + \frac{1}{r}A_\varphi\right)\vec{z} + 0\cdot\vec{\varphi}$$

∂ everything but $\vec{\square}$

13? $\nabla + \nabla\vec{A}$???

$$\begin{pmatrix}\frac{\partial}{\partial r}\\ \frac{\partial}{\partial\varphi}\\ \frac{\partial}{\partial z}\end{pmatrix} + \begin{pmatrix}\frac{\partial}{\partial r}\\ \frac{\partial}{\partial\varphi}\\ \frac{\partial}{\partial z}\end{pmatrix} \quad ?????$$

→ curl again: $\nabla\times\nabla\times\vec{A}$?

$$= -\left((A_\varphi)_{zz} + (A_\varphi)_{rr} + \frac{1}{r}(A_\varphi)_r - \frac{1}{r^2}(A_\varphi)\right)$$

$$\begin{aligned}\nabla^2 A &= \nabla(\nabla\cdot\vec{A}) - \nabla\times\nabla\times\vec{A}\\ &= \nabla(0) + \left((A_\varphi)_{zz} + (A_\varphi)_{rr} + \frac{1}{r}(A_\varphi)_r - \frac{1}{r^2}(A_\varphi)\right)\vec{\varphi}\\ &= \left((A_\varphi)_{zz} + (A_\varphi)_{rr} + \frac{1}{r}(A_\varphi)_r - \frac{1}{r^2}(A_\varphi)\right)\vec{\varphi}\end{aligned}$$

so (1): $\nabla^2\vec{A} + k_0^2\cdot A_\varphi = 0$

$$= \left((A_\varphi)_{zz} + (A_\varphi)_{rr} + \frac{1}{r}(A_\varphi)_r - \frac{1}{r^2}(A_\varphi) + k_0\cdot A_\varphi\right)\vec{\varphi}$$

Assume $A_\varphi = R(r)\cdot Z(z)$, let $A_\varphi = A$

thus
$A_{zz} = R(r)\cdot Z''(z)$
$A_r = Z(z)\, R'(r)$
$A_{rr} = Z(z)\cdot R''(r)$

$$\Big\}\; R Z'' + Z\cdot R'' + \frac{1}{r}\cdot Z\cdot R' - \frac{1}{r^2}RZ + k_0^2\cdot RZ = 0$$

$$R Z'' + Z\cdot R'' + \frac{1}{r}\cdot Z\cdot R' - \frac{1}{r^2}RZ + k_0^2\cdot RZ = 0$$ → basically an ODE

divide by $A = RZ \neq 0$ otherwise trivial solution

$$\frac{1}{Z}\cdot Z'' + \frac{1}{R}\cdot R'' + \frac{1}{r\cdot R}\cdot R' - \frac{1}{r^2} + k_0^2 = 0$$

$$\Leftrightarrow \frac{1}{Z}\cdot Z'' + k_0^2 = -\left(\frac{1}{R}R'' + \frac{1}{rR}R' - \frac{1}{r^2}\right)$$

in r: $\frac{1}{Z}\cdot Z''$ stays constant → set $-\lambda^2$

in z: same : let y

r: $\frac{1}{R}\cdot R'' + \frac{1}{r}\cdot\frac{1}{R}R' - \frac{1}{r^2} = -\lambda^2$

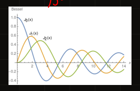
$$R'' + \frac{1}{r}\cdot R' - \frac{1}{r^2}\cdot R = -\lambda^2\cdot R$$

$$\Leftrightarrow R'' + \frac{1}{r}\cdot R' + \left(\lambda^2 - \frac{1}{r^2}\right)R = 0$$ → Bessel : $R = A\cdot J_1(\lambda r) + B\cdot Y_1(\lambda r)$

or $A\cdot J_1(\lambda r)\cdot\begin{cases}B\cdot e(\tau_0 z) & z<0\\ C\cdot e(-\tau_0 z) & z>0\end{cases}$ *

order 1 Bessel

so beautiful and natural



2d kind looks more like some log?

expected pattern

guarantees convergence in z → ∞

z: $\frac{1}{Z}Z'' + k_0^2 = \lambda^2$

$$\Leftrightarrow Z'' = (\lambda^2 - k_0^2)Z$$

$$\Leftrightarrow Z'' - (\lambda^2 - k_0^2)Z = 0$$

$$= Z'' + \underbrace{(k_0^2 - \lambda^2)}_{x}Z = 0$$

$$k^2 + x = 0 \Rightarrow x = \pm\sqrt{x} = \pm\sqrt{k_0^2 - \lambda^2}$$

$$\rightarrow Z = A\cdot e\left(\sqrt{k_0^2-\lambda^2}\cdot z\right) + B\cdot e\left(-\sqrt{k_0^2-\lambda^2}\cdot z\right)$$

$$= A\cdot e(\tau_0 z) + B\cdot e(-\tau_0 z)$$

* No boundaries → r and z → ∞ : continuous → no Σ but ∫

↳ no limit on λ → account for all of them

with $z<0$: $A_\varphi^- = \int_0^{+\infty} B^-(\lambda)\cdot J_1(\lambda r)\cdot e\left(\sqrt{\lambda^2-k_0^2}\cdot z\right)d\lambda$

with $z>0$: $A_\varphi^+ = \int_0^{\infty} B^+(\lambda)\, J_1(\lambda\cdot r)\cdot e\left(-\sqrt{\lambda^2-k_0^2}\cdot z\right)d\lambda$

} (A)

take $z=0$: $A_\varphi^- = \int_0^\infty d\lambda = 0$, $A_\varphi^+ = \int_0^\infty d\lambda = 0$ } $A_\varphi^- = A_\varphi^+$ if $z=0$ → $B^- = B^+$ } in $r\neq 0$

ook. $K = \vec{n}\times(H_2 - H_1)$: $\vec{n}\parallel\vec{z}$ → $H_{1,2}$ are radial

→ like celozzi

↳ intuitive ≡ $\vec{H}\propto\vec{B}$

current follows φ

↓

$\vec{B} = \nabla\times\vec{A}$:(22)

RHS : source → $\vec{\varphi}\cdot I\cdot\delta(r-a) = K$ place current at r = a : ideal magnetic "point"

LHS : $K = \vec{z}\times(\underset{z>0}{H_2} - \underset{z<0}{H_1})$

$$\vec{z}\times(H_2 - H_1) = \vec{\varphi}\cdot I\cdot\delta(r-a)$$

$$= \vec{z}\times\left[(H_{2r}\vec{r} + H_{2z}\vec{z}) - (H_{1r}\vec{r} + H_{1z}\vec{z})\right] = \text{—}$$

$$= \left(H_{2r}\underbrace{\vec{r}\times\vec{z}}_{\vec{\varphi}} + H_{2z}\underbrace{\vec{z}\times\vec{z}}_{0}\right) - \left(H_{1r}\underbrace{\vec{r}\times\vec{z}}_{\vec{\varphi}} + H_{1z}\underbrace{\vec{z}\times\vec{z}}_{0}\right) = \text{—}$$

$$= (H_{2r} - H_{1r})\cancel{\vec{\varphi}} = I\,\delta(r-a)\cancel{\vec{\varphi}}$$

$$= H_{2r} - H_{1r} = I\,\delta(r-a)$$

* $B = \mu_0\cdot H$ en $B_r = -\frac{\partial A_\varphi}{\partial z}\vec{r}$

so: $-\frac{1}{\mu_0}\left(\frac{\partial A_{2\varphi}}{\partial z} - \frac{\partial A_{1\varphi}}{\partial z}\right) = I\cdot\delta(r-a)$

$$\Leftrightarrow \frac{\partial A_{2\varphi}}{\partial z} - \frac{\partial A_{1\varphi}}{\partial z} = -\mu_0\cdot I\cdot\delta(r-a)$$

$$\Leftrightarrow \left.\frac{\partial A_{1\varphi}}{\partial z}\right|_{z=0-\varepsilon} - \left.\frac{\partial A_{2\varphi}}{\partial z}\right|_{z=0+\varepsilon} = \mu_0 I\,\delta(r-a)$$

but $z=0\neq$ continuous → $z = 0\overset{2}{\underset{1}{\pm}}\varepsilon$

↳ $\nabla\cdot B = 0$ Gauss

→ ($\mathcal{B}$ in $0-\varepsilon$) ≡ ($\mathcal{B}$ in $0+\varepsilon$)

$$B_{1z} = B_{2z}$$

Radially: Ampere : $\nabla\times H = J$: sign jumps at $z=0$

$B_z = (\nabla\times A)_z$

$$= \frac{\partial A_\varphi}{\partial r} + \frac{1}{r}A_\varphi = \frac{1}{r}\cdot\frac{\partial}{\partial r}(A_\varphi\cdot r)$$ → still $A_{1\varphi} = A_{2\varphi}$: current in $z=0$ } Tangential

taking the partials in (A) with ε→0

$$\lim_{\varepsilon\to0}\left(\left.\frac{\partial A_{1\varphi}}{\partial z}\right|_{z=0-\varepsilon} - \left.\frac{\partial A_{2\varphi}}{\partial z}\right|_{z=0+\varepsilon}\right) = \mu_0\cdot I\cdot\delta(r-a)$$

$$= \int_0^\infty B(\lambda)\cdot J(\lambda r)\cdot\sqrt{\lambda^2-k_0^2}\,d\lambda - \int_0^\infty B(\lambda)\cdot J_1(\lambda r)\cdot\left(-\sqrt{\lambda^2-k_0^2}\right)d\lambda = \text{—}$$

$$= 2\cdot\int_0^\infty\boxed{B(\lambda)}\, J_1(\lambda r)\sqrt{\lambda^2-k_0^2}\,d\lambda = \mu_0 I\,\delta(r-a)$$

take λ ↦ λ' as dummy and multiply by $J_1(\lambda r)r$ for

$$\int_0^\infty 2\cdot\int_0^\infty B(\lambda')J_1(\lambda' r)\sqrt{\lambda'^2-k_0^2}\,d\lambda'\cdot J_1(\lambda r)r\,dr = \int_0^\infty \text{—}\cdot J_1(\lambda r)r\,dr$$

$$= 2\cdot\left[\int_0^\infty\underset{\perp}{\underline{J_1(\lambda r)r}}\,dr\right]\cdot\left[\int_0^\infty B(\lambda')\underline{J_1(\lambda' r)}\sqrt{\lambda'^2-k_0^2}\,d\lambda'\right] = \mu_0\cdot I\cdot\int_0^\infty\delta(r-a)\cdot J_1(\lambda r)\cdot r\,dr$$

→ some theorem : $B(\lambda) = \frac{\mu_0\cdot a\cdot I}{2}\cdot\frac{\lambda}{\sqrt{\lambda^2-k_0^2}}\cdot J_1(\lambda a)$

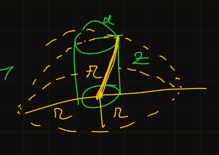
↓ (19, 20)

$A_{1\varphi} = \frac{\mu_0\cdot I\cdot a}{2}\int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot e(\tau_0 z)\,d\lambda$, where $z\in\mathbb{Z}_0^-$

$A_{2\varphi} = (\;\;"\;\;)\int_0^\infty "\cdot"\cdot"\cdot e(-\tau_0 z)\,d\lambda$, where $z\in\mathbb{Z}_0^+$

Sommerfeld : $\int_0^\infty\frac{\lambda}{\tau}\cdot J_0(\lambda a)\,e(-\tau_0 z)\,d\lambda = \frac{e(-jk_0 r)}{r}$

λ; $\sqrt{\lambda^2-k^2}$; r(loop); $\sqrt{a^2+z^2}$



connects coordinates cylindrical = spatial

cylindrical J_0 waves ≡ spherical waves

using $\left(J_0(\lambda a)\right)'_a = -\lambda J_1(\lambda a)$

cos ↳ zeroes → heigth

$$\frac{\partial A_\varphi}{\partial a} = -\int_0^\infty\frac{\lambda^2}{r}J_1(\lambda a)\,e(-\tau_0 z)\,d\lambda$$

↳ relates to $B_z = \frac{\mu_0 a I}{2}\left(jk_0 a + \frac{a}{r}\right)\frac{e(-jk_0 r)}{r^2}$

small a, low f and far out

→ $B_z \approx \frac{\mu_0 a^2 I}{2\sqrt{r^3}}$ γ

$$A_\varphi = A_{\varphi(in)} + A_{\varphi(ind)}$$

$$A_{\varphi(1)} = \gamma\left\{\int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot\overbrace{e(-\tau_0|z|)}^{\text{incoming}}\,d\lambda + \int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot\overbrace{C(\lambda)\cdot e(-\tau_0 z)}^{\text{induced + reflections}}\,d\lambda\right\}$$] source & reflections

$$A_{\varphi(2)} = \gamma\left\{\int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot e(-\tau_0 z)\cdot\overrightarrow{D}(\lambda)\,d\lambda + \int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot\overleftarrow{E}(\lambda)\cdot e(\tau_0 z)\,d\lambda\right\}$$] within shield

$\tau_0 = \sqrt{\lambda^2 - k_0^2}$ ↳ $j\omega\mu\sigma$

$$A_{\varphi(3)} = \gamma\int_0^\infty\frac{\lambda}{\tau_0}J_1(\lambda a)\cdot J_1(\lambda r)\cdot F(\lambda)\cdot e(-\tau_0 z)\,dz$$

in z = b : border (1) & (2) → $\mu_0\to\mu$

at this border $A_{\varphi(1)} = A_{\varphi(2)}$

$$\frac{1}{\mu_0}\frac{\partial A_{\varphi(1)}}{\partial z} = \frac{1}{\mu}\frac{\partial A_{\varphi(2)}}{\partial z}$$

in z = c : border (2) & (3)

" " " (2) " " (3)

$$\frac{1}{\mu}\frac{\partial A_{\varphi(2)}}{\partial z} = \frac{1}{\mu_0}\frac{\partial A_{\varphi(3)}}{\partial z}$$

i believe his math

Matrix composition

either side of the shield has jump conditions *

→ same kernel $= m(\lambda, r) = \gamma \cdot \frac{\lambda}{\gamma_0} J_1(\lambda a) J(\lambda r)$ at $1 \to 2$ & $2 \to 3$

→ z dependent parts differ → divide $M(\lambda, r)$ out

$$\left.\begin{aligned} 1&: e(-\gamma_0 z) + C \cdot e(\gamma_0 z) \\ 2&: D \cdot e(-\gamma z) + E\, e(\gamma z) \\ 3&: F \cdot e(-\gamma_0 z) \end{aligned}\right\} \text{fields}$$

$A_{\varphi 1}(z=b) = A_{\varphi 2}(z=b)$ & $A_{\varphi 2}(z=c) = A_{\varphi 3}(z=c) \Rightarrow$ continuity A_φ

and $H_r = -\frac{1}{\mu} \cdot \frac{\partial A_\varphi}{\partial z} \Rightarrow A'_{\varphi 1}(z=b) = A'_{\varphi 2}(z=b)$ & $A'_{\varphi 2}(z=c) = A'_{\varphi 3}(z=c) \Rightarrow$ continuity H_r

$z = b$

A_φ: $e(-\gamma_0 b) + C \cdot e(\gamma_0 b) = D \cdot e(-\gamma b) + E \cdot e(\gamma b)$

$\Leftrightarrow C \cdot e(\gamma_0 b) - D \cdot e(-\gamma b) - E \cdot e(\gamma b) = -e(\gamma_0 b)$ (1)

H_r: $\frac{\gamma_0}{\mu_0} \cdot C \cdot e(\gamma_0 b) + \frac{\gamma}{\mu} \cdot D \cdot e(-\gamma b) - \frac{\gamma}{\mu} \cdot E \cdot e(\gamma b) = \frac{\gamma_0}{\mu_0} e(-\gamma_0 b)$ (2)

$z = c$

A_φ: $D \cdot e(-\gamma c) + E \cdot e(\gamma c) = F\, e(-\gamma_0 c)$

$\Leftrightarrow D \cdot e(-\gamma c) + E \cdot e(\gamma c) - F \cdot e(-\gamma_0 c) = 0$ (3)

H_r: $\frac{\gamma}{\mu} \cdot E \cdot e(\gamma c) - \frac{\gamma}{\mu} \cdot D \cdot e(-\gamma c) + \frac{\gamma_0}{\mu_0} \cdot F \cdot e(-\gamma_0 c) = 0$ (4)

matrix

$$\begin{bmatrix} e(\gamma_0 b) & -e(\gamma b) & -e(\gamma b) & 0 \\ \vdots & & & \\ \vdots & & & \end{bmatrix} \begin{bmatrix} C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} -e(-\gamma_0 b) \\ \frac{\gamma_0}{\mu_0} e(-\gamma_0 b) \\ 0 \\ 0 \end{bmatrix}$$

& K defined as $\frac{\gamma}{\gamma_0}$ and $\mu_r = \frac{\mu}{\mu_0}$

$t = c - b$

trust the algebra

i am not solving this system
see moser (39-42) for the solutions
this gives

$A_\varphi(1) = \gamma \cdot \int_0^\infty d\lambda \cdot \left\{ \frac{\lambda}{\gamma_0} J_1(\lambda a) J_1(\lambda r) \cdot \left[e(-\gamma_0 |z|) + C \cdot e(\gamma_0(z - 2b)) (K^2 - \mu_r^2)(e(-2t\gamma) - 1) \right] \right\}$

$A_\varphi(2) = 2 \cdot \gamma \cdot \int_0^\infty d\lambda \cdot \left\{ \frac{\lambda}{\gamma_0} J_1(\lambda a) J_1(\lambda r) \cdot C \cdot e(b(\gamma - \gamma_0)) \cdot \left[e(-\gamma z)(K + \mu_r) + e(\gamma(z - 2c)) \cdot (K - \mu_r) \right] \right\}$

$A_\varphi(3) = 4 \cdot \gamma \cdot \int_0^\infty d\lambda \cdot \left\{ C \cdot K \cdot \lambda \cdot J_1(\lambda a) \cdot J_1(\lambda r) \cdot e(-\gamma_0 z - t(\gamma - \gamma_0)) \right\}$

(10, 12)

$$\vec{B}_1 = \nabla \times \vec{A}_1 = \frac{\mu_0 a I}{2} \cdot \left\{ \vec{r} \int_0^\infty \lambda J_1(\lambda a) J_1(\lambda r) \Big(e(-\gamma_0 z) - C \cdot e(\gamma_0(z - 2b)) \cdot (K^2 - \mu_r^2)(e(-2tr) - 1) \Big) d\lambda \right.$$
$$\left. + \vec{z} \cdot \int_0^\infty \frac{\lambda^2}{\gamma_0} J_0(\lambda r) \cdot J_1(\lambda a) \cdot \Big(e(-\gamma_0 |z|) + C\, e(\gamma_0(z - 2b)) \cdot (K^2 - \mu_r^2)(e(-2tr) - 1) \Big) d\lambda \right\}$$

$E \perp B$: $E = -j\omega \vec{A} - \nabla \Phi$

$= -j\omega \vec{A} + \frac{\nabla(\cancel{\nabla \cdot A})}{j\omega\mu\varepsilon} \rightarrow \phi$ (11)

$= -j\omega \vec{A}$

$\vec{E}_1 = -j\omega A_1$

$= -j\omega\gamma \int_0^\infty d\lambda \left\{ \frac{\lambda}{\gamma_0} J_1(\lambda a) J_1(\lambda r) \left[e(-\gamma_0 |z|) + C \cdot e(\gamma_0(z - 2b)) (K^2 - \mu_r^2)(e(-2tr) - 1) \right] \right\} \vec{\varphi}$

$$\vec{B}_2 = 2 \left\{ \int_0^\infty d\lambda \left\{ M(\lambda, r) \cdot C \cdot \overset{\text{waves}}{e}(\overset{\text{jump}}{b}(\gamma - \gamma_0)) \left[\overbrace{e(-\gamma z)(K + \mu_r)}^{\text{into}} - \overbrace{e(\gamma(z - 2c))(K - \mu_r)}^{\text{out of}} \right] \right\} \vec{r} \right.$$
$$\left. + \int_0^\infty d\lambda \left\{ \frac{\lambda^2}{\gamma_0} \cdot J_1(\lambda a) J_0(\lambda r) \cdot C \cdot e(b(\gamma - \gamma_0)) \left[e(-\gamma z)(K + \mu_r) + e(\gamma_0(z - 2c))(K - \mu_r) \right] \right\} \vec{z} \right\}$$

im not writing the rest B_3 is needed for S

$$\vec{B} = \nabla \times \vec{A} = \gamma \left\{ \int_0^\infty d\lambda \left\{ \gamma_0 \cdot M(\lambda, r) \cdot e(-\gamma_0 z) \right\} \vec{r} + \int_0^\infty d\lambda \left\{ \frac{\lambda^2}{\gamma_0} J_1(\lambda a) J_0(\lambda r) \cdot e(-\gamma_0 z) \right\} \vec{z} \right\}$$

then $S := 20 \cdot \log \left| \frac{B_z}{B_{z(3)}} \right|$ · high S ⇒ good shield

$$= 20 \cdot \log \left\{ \frac{1}{4\mu_r} \cdot \left| \frac{\int_0^\infty d\lambda \left\{ \frac{\lambda^2}{\gamma_0} J_1(\lambda a) \cdot e(-\gamma_0 z) \right\}}{\int_0^\infty d\lambda \left\{ \frac{C \lambda^2 \gamma}{\gamma_0^2} \cdot J_1(\lambda a)\, e(-\gamma_0 z - t(\gamma - \gamma_0)) \right\}} \right| \right\} \longrightarrow \text{needs an analytical solution}$$

simplifies by (32) ↳ a small, z big, ω low ⇒ λ high

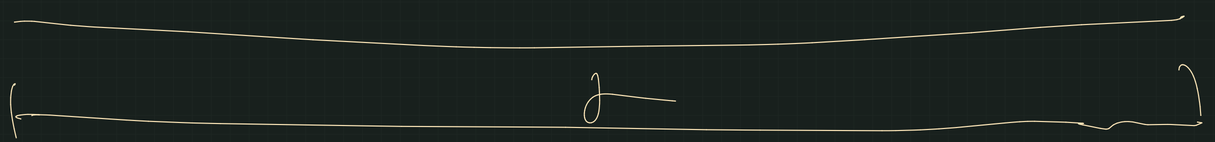
$$\approx -20 \cdot \log \left\{ \frac{4 \cdot \mu_r}{a} \cdot \sqrt{r^3} \cdot \left| \int_0^\infty d\lambda \left\{ \frac{C \cdot \lambda^2 \gamma}{\gamma_0^2} \cdot J_1(\lambda a)\, e(-\gamma_0 z - t(\gamma - \gamma_0)) \right\} \right| \right\}$$

$$Z_P = \frac{\Phi}{I}$$

$$f_s \approx \frac{1}{\sqrt{LC}}$$

→ Φ ~~[illegible]~~ Long

$L \approx$ length



$f_{res} = \frac{\omega_{res}}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$

$L = \frac{\mu \cdot N^2 \cdot A}{l}$: $N\uparrow \Rightarrow L\uparrow\uparrow \Rightarrow f_{res}\downarrow\downarrow$

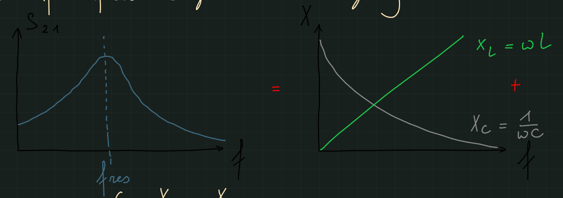
$C \sim X_C = \frac{1}{\omega C}$: $C\downarrow \Rightarrow X_C\uparrow \Rightarrow f_{res}\downarrow$

↳ depends on N : $C = \frac{\varepsilon \cdot A}{d}$

think s parameters or impedance

at $f > f_{res}$: inductive coupling → magnetic behaviour

at $f < f_{res}$: capacitive coupling → electric behaviour



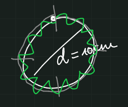
↳ $X_L = X_C$

$\omega L = \frac{1}{\omega C}$

→ high $f \Rightarrow$ with small $\sqrt{LC} \Rightarrow$ L or C very small

⇒ this is why $r = \sqrt{a^2 + z^2}$ is needed : infinitely tiny loop is not possible or could be capacitive

if l (= rolled out length) $\ll \lambda = c/f$ we won't measure $\Delta\varphi$



f order 10 MHz : λ order 30 m

→ no $\Delta\varphi$: change in A way too small

f order 10 GHz : λ order 3 cm

→ measurable $\Delta\varphi$

1) Generalize to $x, y \neq 0 \Rightarrow$ positions within a plane at z

$A = A_\varphi \cdot \vec{\varphi}$

before shield : $A_\varphi^1 = \underline{\mu_0 \cdot a \cdot I} \int_0^\infty d\lambda \left\{ \left[J_1(\lambda a) J_1(\lambda \rho) \cdot \left[e(-\tau_0|z|) + C \cdot e(\tau_0(z - 2 \cdot b)) \right] (K^2 - \mu_r^2)(e(-2t\tau) - 1) \right] \right\}$, stays

inside shield : $A_\varphi^2 = \mu \cdot a \cdot 2 \cdot I \cdot \int_0^\infty d\lambda \left\{ \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) \cdot C \cdot e(b(\tau - \tau_0)) \cdot \left[e(-\tau \cdot z) \cdot (K + \mu_r) + e(\tau(z - 2 \cdot c)) \cdot (K - \mu_r) \right] \right\}$, stays

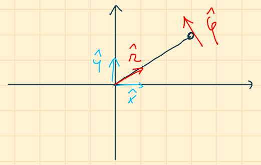
$\tau = \sqrt{\lambda^2 - k^2}$ $\quad \frac{\tau}{\tau_0}$ $\quad [(K + \mu_r)^2 - (K - \mu_r)^2 e(-2t\tau)]^{-1}$

after schield $A_\varphi^3(r, z) = 2\mu a \cdot I \cdot \int_0^\infty d\lambda \left\{ \frac{C \cdot \lambda \cdot \tau}{\tau_0^2} \cdot J_1(\lambda a) \cdot J_1(\lambda r) \cdot e(-\tau_0 z - t(\tau - \tau_0)) \right\}$, $r \mapsto \sqrt{(x - x_0)^2 + (y - y_0)^2} = \sqrt{x^2 + y^2}$

$$\vec{A}^3(x, y, z) = A_\varphi^3(r, z) \cdot \hat{\varphi}$$

$$= A_\varphi^3(x, y, z) \cdot \hat{\varphi} = 2 \cdot \mu \cdot a \cdot I \cdot \int_0^\infty d\lambda \left\{ \frac{C \lambda \tau}{\tau_0^2} J_1(\lambda a) J_1(\lambda \sqrt{x^2 + y^2}) \cdot e(-\tau_0 z - t(\tau - \tau_0)) \right\}$$

$\hat{\varphi} \rightarrow$ carthesian



$$\hat{r} = \cos(\varphi) \cdot \hat{x} + \sin(\varphi)\hat{y}$$

$$= \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y}$$

$$\hat{r} \perp \hat{\varphi} \Rightarrow \hat{\varphi} = -\sin(\varphi) \cdot \hat{x} + \cos(\varphi) \cdot \hat{y}$$

$$= -\frac{y}{r} \cdot \hat{x} + \frac{x}{r}\hat{y}$$

$$= -\frac{y}{\sqrt{x^2 + y^2}} \hat{x} + \frac{x}{\sqrt{x^2 + y^2}} \hat{y}$$

so $\vec{A}^3(x, y, z) = A_\varphi^3(x, y, z)\ \hat{\varphi}$

$$= A_\varphi^3(x, y, z) \cdot \left[\frac{-y}{\sqrt{x^2 + y^2}} \hat{x} + \frac{x}{\sqrt{x^2 + y^2}} \hat{y} \right]$$

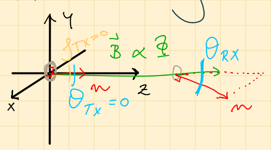
$\Rightarrow \vec{A}_x^3(x, y, z) = \frac{-y}{\sqrt{x^2 + y^2}} \cdot \vec{A}_\varphi^3(x, y, z)$ and $\vec{A}_y^3(x, y, z) = \frac{x}{\sqrt{x^2 + y^2}} \cdot \vec{A}_\varphi^3(x, y, z)$

Matlab looks ok

→ not much visual difference when adding / removin shield

2) Generalize to $\Phi(\Theta)$

A) R tiny . TX & RX : vectors → dot product



$\theta : \angle(x_0, z)$

$\varphi : \angle(y_0, z)$

$\vec{B}$: perpendicular field vector

r : orientation

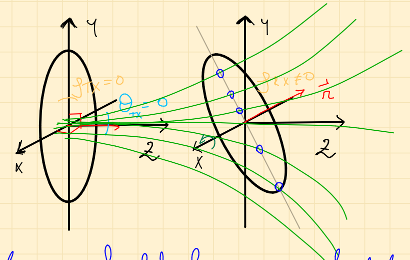
$\Phi = \int_{rx} \vec{A}_{LTX} \cdot d\vec{l} \approx (\vec{B}(\vec{c}) \cdot \hat{n}) \cdot A$ → basically constant over A

* TX loop parametrization : $\vec{p}(\theta) = \vec{c} + R \cdot (\cos(\theta)\hat{u} + \sin(\theta)\hat{v})$

$d\vec{l} = d\vec{p} = R(\cos(\theta)\hat{u} - \sin(\theta)\hat{v})\,d\theta$

$$\Phi(\theta) = \int_{rx} d\theta \cdot \{ \vec{A}_{LTX}(\vec{p}(\theta)) \cdot R(\cos(\theta)\hat{u} - \sin(\theta)\hat{v}) \}$$

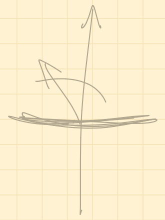
B) rx kinda big : difference in Φ through $A(RX)$ → line or surface integral



$\theta : \angle(x_0, z)$

$\varphi : \angle(y_0, z)$

$\vec{r}$: orientation



lower half has one field line less } → full line integral over RX
upper half has one field line more }

* again $\vec{p}(\theta) = \vec{c} + R \cdot (\cos(\theta)\hat{u} + \sin(\theta)\hat{v})$

when $\vec{c}, \hat{u}, \hat{v} \in \mathbb{R}^3$ and turned about x

$\hat{u} = \hat{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $R_x(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$

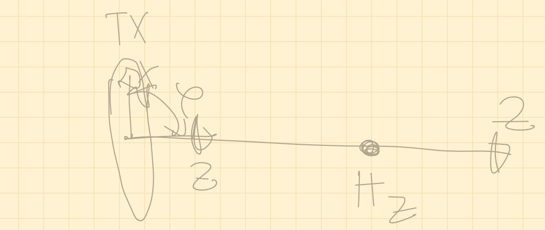
$\hat{v} = \hat{y} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $R_y(\varphi) = \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{pmatrix}$

$$R(\theta, \varphi) = R_x(\theta) \cdot R_y(\varphi) = \begin{pmatrix} \cos(\theta) \cdot \cos(\varphi) & -\sin(\theta) & \cos(\theta)\sin(\varphi) \\ \sin(\theta)\cos(\varphi) & \cos(\theta) & \sin(a)\sin(\varphi) \\ -\sin(\varphi) & 0 & \cos(\varphi) \end{pmatrix}$$

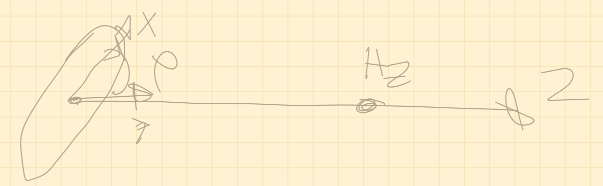
$\hat{u}_R = R(\theta, \varphi) \cdot \hat{u} = (c(\theta)c(\varphi), s(\theta)c(\varphi), -s(\varphi))^T$ when turning about y first
$\hat{v}_R = R(\theta, \varphi) \cdot \hat{v} = (-s(\theta), c(\theta), 0)^T$
$\hat{n}_R = R(\theta, \varphi) \cdot \hat{v} = (c(\theta)s(\varphi), s(\theta)s(\varphi), \cos(\varphi))^T$

so $\vec{p}(\theta, \varphi) = \vec{c} + R \cdot \cos(\varphi) \begin{pmatrix} \cos(\theta)\cos(\varphi) \\ \sin(\theta)\cos(\varphi) \\ -\sin(\varphi) \end{pmatrix} + R \sin(\varphi) \begin{pmatrix} -\sin(\theta) \\ \cos(\theta) \\ 0 \end{pmatrix}$ * full loop * about x * about y

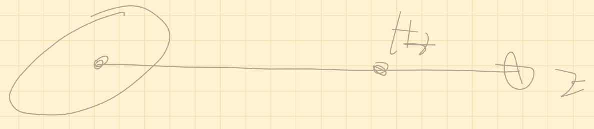
→ basically the same as R tiny, at least generalized to both θ and φ



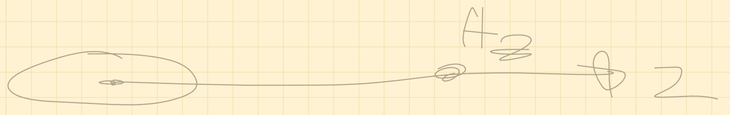
(1) $\varphi = 90°$



(2) $\varphi = 60°$



(3)



(4)

$SE = \frac{H_{close}}{H_{open}}$

[illegible]

Aranco / Moser

* configuration

- M · parallel · allows "projection" to cilindrical 2D wave ⟹ Bessel
- A : coplanar · full 3D problem

* Assumptions

- M : $R \to 0$: infinitely small loop ⟹ solution for points in 2D space at $d = x$
- Aranco : finite sized loop : surface / contour integration needed ⟹ $\bar{B}$ distribution

- M · low frequency range : $k << R, d, \ldots$ wave number does nothing
- A : $e(-j \cdot k \cdot R) \approx 1$ at $R = 1'$ & $f = 16$ MHz : $\lambda >> R, d, \ldots$

- M : $SE \propto B / B_s$
- A : $SE \propto \Phi / \Phi_s$

* Starting point

- M : 1st order vector potential $\bar{A}$: only A_φ in CC
- A : 2nd " " " $\bar{A} = \nabla \times \bar{U}$, $\nabla \times (\nabla \times \bar{U}) = \bar{B}$: 3 components ·

axial

$$B_x = \frac{\partial^2 U_{\textcircled{y}}}{\partial x \partial y}$$
$$B_y = \partial^2 U_y / \partial y^2$$
$$B_z = \partial^2 U_y / \partial y \partial z$$

→ this cartesian domain leads to $J_1 \to I_1$: cartesian Bessel

* only the kernel changes bc of the shield

$$G_0(\alpha, \beta) = -\frac{\mu_0 \cdot I \cdot R}{\pi} \cdot \frac{1}{\beta \cdot \sqrt{\alpha^2 + \beta^2}} \cdot I_1(\beta R)$$

↓ shield

$G_1(\alpha, \beta)$ just look at the paper

$$B_y = \frac{\partial^2}{\partial y^2} \int_0^\infty \int_0^\infty d\alpha \, d\beta \left\{ G_x(\alpha, \beta) \cdot \cos(\alpha \cdot x) \cdot \underline{\cos(\beta \cdot y)} \cdot e(-D \cdot |z + d|) \right\}$$

$$= -\int_0^\infty \int_0^\infty d\alpha \, d\beta \left\{ \beta^2 \, G_x(\alpha, \beta) \cdot \cos(\alpha \cdot x) \cdot \cos(\beta y) \cdot e(-D \cdot |z + d|) \right)$$

* $G_x(\alpha, \beta)$: loop
→ I, R, bessel

* $\cos(\alpha x)$ & $\cos(\beta y)$: spatial "waves" (?)

* $e(D|z+d|)$: propagation in z

* α, β : spatial frequencies
→ all f like the top hat in cilindrical

○ Approximation : $\iint_{RX} B_y \, dS \approx S_{RX} \cdot \sum w_i \cdot B_y(x_i, z_i)$

total flux ≈ weighted B_y points